

ANOVA → Analysis of Variance

H_0 : mean accuracy of RF = mean accuracy of XGBoost = mean of sum

H_1 : at least one mean is different

F-Statistics

Variation b/w groups
70, 80, 90

Variation within groups
70, 80, 90
71, 72, 73
69, 70, 71
very little.

If $F = 1$
No significant difference
groups are similar
fail to reject H_0

If F is very large
 $F > 25$
strong evidence against H_0

If F is moderately large
 $F \approx \{4 \text{ to } 8\}$
groups may be different
need p-value confirmation

$\alpha = 0.05$

$\underline{1} \text{ If } p < \alpha$
reject H_0

$\underline{2} \text{ If } p \geq \alpha$
fail to reject H_0

$F_{stat} = 12$
 $p\text{-value} = 0.002$
 $\alpha = 0.05$

$0.002 < 0.05$
reject H_0

$F_{stat} = 1.5$
 $p\text{-value} = 0.38$
 $\alpha = 0.05$

$0.38 > 0.05$
fail to reject H_0

A	B	C
70	80	90
72	82	92
68	78	88

$\bar{x}_A = 70$ $\bar{x}_B = 80$ $\bar{x}_C = 90$

Grand mean = $\frac{70+72+68+80+82+78+90+92+88}{9} = 80$

SSB (Sum of Square b/w Groups)

$SS_B = \sum n_i (\bar{x}_i - \bar{x})^2$

for $C_A = 3(70 - 80)^2 = 300$

$C_B = 3(80 - 80)^2 = 0$

$C_C = 3(90 - 80)^2 = 300$

$= 300 + 0 + 300 = 600$

SSW (Sum of squares within group)

$SS_W = \sum (x - \bar{x})^2$

$\begin{pmatrix} 70 \\ 72 \\ 68 \end{pmatrix} C_A = (70-70)^2 + (72-70)^2 + (68-70)^2 = 8$

$\begin{pmatrix} 80 \\ 82 \\ 78 \end{pmatrix} C_B = (80-80)^2 + (82-80)^2 + (78-80)^2 = 8$

$\begin{pmatrix} 90 \\ 92 \\ 88 \end{pmatrix} C_C = (90-90)^2 + (92-90)^2 + (88-90)^2 = 8$

$SS_W = 8 + 8 + 8 = 24$

Degree of freedom df

Between groups $df_B = k - 1 = 3 - 1 = 2$

Within group $df_W = N - k = 9 - 3 = 6$

Mean squares

Between = $\frac{SS_B}{df_B} = \frac{600}{2} = 300$

Within = $\frac{SS_W}{df_W} = \frac{24}{6} = 4$

F-statistic = $\frac{MS_B}{MS_W} = \frac{300}{4} = 75$

CHI-Square test χ^2

$p\text{-value} < \alpha$
reject H_0

Small χ^2 → observed data is close to expected data → independent

large χ^2 → ———— differs from ———— → associated

Gender	Purchased	Not purchased	Total
Male	30	20	50
Female	20	40	50
Total	40	60	

$E = \frac{(\text{Row total})(\text{Column total})}{GT}$

for male: $\frac{50 \times 40}{100} = 20$

→ is purchase behaviour dependent on gender?

Fair die → 60 times

6 sides

Side	Observed	Expected
1	8	10
2	12	10
3	10	10
4	10	10
5	13	10
6	7	10

$\frac{60}{6} = 10$ for

3 numbers → 10

Sum → $10 \times 3 = 30$

$8 + 12 + x = 30$
 $x = 10$

$df = n - 1$

$a = \frac{\text{sum}}{\text{number}}$
 $\text{sum} = \text{av} \times \text{number}$

$150 + 20 + x = 30$
 $x = -140$

$3 - 1 = 2$

out of 3 values, only 2 are free to vary.

Statistical Visualization

Employee	Salary
A	30000
B	35000
C	40000
D	45000
E	50000

Visualization toolkit

- Histogram
- Box plot
- Bar plot
- Q-Q plot
- Probability Plot
- Residual plot

Skewed

- symmetric
- positive skew → mean > median > mode
- negative skew → mean < median < mode

$S = \frac{3(\text{Mean} - \text{Median})}{Std}$

↓

Pearson

- 0 → perfect normal
- 0.5 to 0.5 → approx symmetric
- 0.5 to 1 → moderately skewed
- > 1 → highly skewed